1 Gradient $m = -3$ **[B1]**

$5 = -3(-2) + c$ or $y - 5 = -3(x - (-2))$ **[M1]**

$y = -3x - 1$ **[A1]**

[Total Marks: 3]

2 **(a)** $\dfrac{-1-5}{5-(-3)}$ or $\dfrac{5-(-1)}{-3-5}$ **[M1]**

$-\dfrac{3}{4}$ **[A1]**

(b) $m = -\dfrac{3}{4}$ **[M1]**

$-2 = -\dfrac{3}{4}(4) + c$ or $y - (-2) = -\dfrac{3}{4}(x - 4)$ **[M1]**

$y = -\dfrac{3}{4}x + 1$ **[A1]**

[Total Marks: 5]

3 Identifies the gradient $m = -2$ and sets up $4 = -2(3) + c$ or $y - 4 = -2(x - 3)$ **[M1]**

$-2x + 10$ **[A1]**

[Total Marks: 2]

4 **(a)** Gradient of $AB = \dfrac{4-2}{7-1} = \dfrac{1}{3}$ **[M1]**

Substitute $(10, 9)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 9 = \dfrac{1}{3}(x - 10)$ **[M1]**

$y = \dfrac{1}{3}x + \dfrac{17}{3}$ or $3y = x + 17$ **[A1]**

(b) Gradient of $BC = \dfrac{9-4}{10-7} = \dfrac{5}{3}$ **[M1]**

Substitute $(1, 2)$ into $y - y_1 = m(x - x_1)$ with their gradient: $y - 2 = \dfrac{5}{3}(x - 1)$ **[M1]**

$y = \dfrac{5}{3}x + \dfrac{1}{3}$ or $3y = 5x + 1$ **[A1]**

(c) Equate equations from (a) and (b): $\dfrac{1}{3}x + \dfrac{17}{3} = \dfrac{5}{3}x + \dfrac{1}{3}$ **[M1]**

$4x = 16 \implies x = 4$ **[A1]**

$y = 7$ **[A1]**

[Total Marks: 9]

5 Rearranges $5x - 2y = 15$ to find the gradient, e.g. $2y = 5x - 15$

y $= \frac{5}{2}x - 7.5$, so $m = \frac{5}{2}$ [M1]

Substitutes $(-4, 1)$ and $m = \frac{5}{2}$ into $y = mx + c$: $1 = \frac{5}{2}(-4) + c$ **[M1]**

$\frac{5}{2}x + 11$ or $2.5x + 11$ **[A1]**

[Total Marks: 3]

6 (a) Calculates the gradient of L_1: $m = \frac{-3-5}{3-(-1)}$ **[M1]**

$m = \frac{-8}{4} = -2$ **[A1]**

Uses parallel gradient -2 and point $C(2, 7)$ to find c: $7 = -2(2) + c$ **[M1]**

$-2x + 11$ **[A1]**

(b) Finds y-intercept $E(0, 11)$ and sets $y = 0$ to find x-intercept D: $0 = -2x + 11$

x $= 5.5[M1]$

Sets up area calculation for right-angled triangle: $\frac{1}{2} \times 5.5 \times 11$ **[M1]**

30.25 or $\frac{121}{4}$ **[A1]**

[Total Marks: 7]